Merge Overlapping Intervals

AlgoExpert · Medium · Arrays

The problem

- Merge all overlapping [start, end] intervals; result sorted by start.

Intuition

- After sorting by start, overlapping intervals are adjacent, so one pass can merge them.

Approach

- 1. Sort by start
- 2. Keep the current merged interval
- 3. Extend its end on overlap
- 4. Otherwise start a new one

Java solution

```java
public class MergeOverlappingIntervals {
    public static int[][] mergeOverlappingIntervals(int[][] intervals) {
        int[][] s = intervals.clone();
        Arrays.sort(s, (x, y) -> Integer.compare(x[0], y[0]));
        List<int[]> out = new ArrayList<>();
        for (int[] iv : s) {
            if (!out.isEmpty() && out.get(out.size() - 1)[1] >= iv[0])
                out.get(out.size() - 1)[1] = Math.max(out.get(out.size() - 1)[1], iv[1]);
            else out.add(new int[]{iv[0], iv[1]});
        }
        return out.toArray(new int[0][]);
    }
}
```

Complexity

- Time: O(n log n)
- Space: O(n)

Recap

- After sorting by start, overlapping intervals are adjacent, so one pass can merge them.
- Complexity: O(n log n)